

TECHNICAL DOCUMENTATION: TOP DRIVE SYSTEM

IDENTITY INFORMATION

****Company_Name:** Meridian Offshore Drilling Solutions LLC**

****Field_Engineer:** James R. Patterson**

****Field_Engineer_Id:** MOS-ENG-4782**

****Document_Version:** 2.3.1**

****Machine_Name:** Automated Top Drive Assembly Unit (ATDAU-7000)**

****Machine_Id:** ATDAU-7000-GOM-2847**

****Installation_Date:** 2023-06-15**

****Number_Of_Units_In_Machine:** 3**

UNIT 1: PRIMARY DRIVE MOTOR

****Unit_Name:** Primary Drive Motor Assembly**

****Unit_Id:** PDM-7000-A1**

****Manufacture_Date:** 2023-04-22**

****Manufacturer_Company_Name:** TorqueMax Industrial Systems Inc.**

****Manufacturer_Company_Id:** TMIS-MFG-5541**

****Inventory_Quantity:** 2 units in stock**

****Supplier_Information:** TorqueMax Industrial Systems Inc., Houston TX; Contact: procurement@torquemaxind.com; Lead Time: 14 business days**

DOMAIN SPECIFICATIONS

- ****Wellbore_Depth_Rating:** 15,000 feet (4,572 meters)**

- ****Operating_Pressure_PSI:** 5,000 PSI maximum continuous**

- ****Drilling_Fluid_Compatibility:** Water-based, synthetic, and oil-based muds (API RP 13B-1 compliant)**

- ****Known_Error_Codes:** E-401 (Motor Overheat), E-402 (Bearing Friction Alert), E-403 (Torque Limiter Engaged)**

- ****Replacement_Part_Number:** PDM-7000-A1-REP**

- ****Maintenance_Schedule:** 500-hour intervals; annual full inspection per API RP 54 Section 8.2**

SECTION 1 - USE CASE

- Primary rotational power source for drill string rotation during vertical and deviated well drilling operations

- Delivers continuous torque output (450-650 ft-lbs) to kelly or kelly-less drilling systems

- Integrates with automated control systems for real-time RPM and torque monitoring
- Supports drilling operations from spud to total depth across multiple well types
- Enables precise speed control (0-200 RPM) for formation-specific drilling optimization

SECTION 2 - FAILURE CAUSES

- ****E-401 (Motor Overheat)**** – Inadequate cooling fluid circulation; frequency: 23% of failures; caused by blocked coolant lines or pump malfunction
- ****E-402 (Bearing Friction Alert)**** – Bearing wear from extended operation without lubrication; frequency: 18% of failures; indicates 200+ hour service interval overdue
- ****E-403 (Torque Limiter Engaged)**** – Excessive downhole resistance or stuck pipe conditions; frequency: 31% of failures; protective shutdown mechanism activated
- ****Seal Degradation**** – Drilling fluid contamination or thermal cycling; frequency: 15% of failures; leads to internal leakage
- ****Electrical Connector Corrosion**** – Saltwater exposure in offshore environments; frequency: 13% of failures; intermittent power loss

SECTION 3 - SOLUTIONS

- ****E-401 Response:**** Immediately cease drilling operations; inspect coolant circulation system per OSHA 1910.147 (Lockout/Tagout); flush cooling lines; verify pump operation; resume only after temperature stabilizes below 180°F
- ****E-402 Response:**** Execute emergency shutdown; perform bearing inspection per API RP 54 Section 9.1; apply ISO VG 32 synthetic lubricant; replace bearings if radial play exceeds 0.015 inches
- ****E-403 Response:**** Reduce weight-on-bit by 15-20%; check for stuck pipe using overpull test (API RP 54 Section 7.3); if unresolved, execute controlled pipe recovery procedure with wireline backup
- ****Seal Replacement Protocol:**** Drain all drilling fluid; disassemble motor housing per manufacturer manual; replace elastomer seals with API-certified equivalents; pressure test at 1.5x operating pressure before resuming operations
- ****Connector Maintenance:**** Apply dielectric grease to all electrical connections quarterly; inspect for green corrosion weekly in offshore operations; replace corroded connectors immediately per OSHA 1910.303 electrical safety standards

SECTION 4 - INVENTORY DATA

- ****Current Stock:**** 2 complete PDM-7000-A1 units; 8 replacement bearing assemblies; 12 seal kits
- ****Reorder Threshold:**** Maintain minimum 1 complete unit and 6 seal kits; trigger reorder when stock reaches threshold
- ****Supplier:**** TorqueMax Industrial Systems Inc., Houston TX; secondary supplier: DrillTech Components Ltd., New Orleans LA (lead time 21 days)
- ****Lead Time & Cost:**** Primary supplier 14 business days; cost per unit \$47,500 USD; expedited shipping available (+\$3,200)

UNIT 2: KELLY DRIVE BUSHING ASSEMBLY

****Unit_Name:** Kelly Drive Bushing with Roller Bearings**

****Unit_Id:** KDB-7000-B2**

****Manufacture_Date:** 2023-05-08**

****Manufacturer_Company_Name:** PrecisionBore Manufacturing Corp.**

****Manufacturer_Company_Id:** PBM-CORP-7823**

****Inventory_Quantity:** 4 units in stock**

****Supplier_Information:** PrecisionBore Manufacturing Corp., Corpus Christi TX; Contact: sales@precisionbore.com; Lead Time: 10 business days**

DOMAIN SPECIFICATIONS

- ****Wellbore_Depth_Rating:** 18,000 feet (5,486 meters)**
- ****Operating_Pressure_PSI:** 3,500 PSI maximum continuous**
- ****Drilling_Fluid_Compatibility:** All standard drilling fluids; compatible with high-temperature high-pressure (HTHP) applications up to 350°F**
- ****Known_Error_Codes:** E-501 (Bushing Wear Detected), E-502 (Roller Bearing Seizure), E-503 (Lateral Play Excessive)**
- ****Replacement_Part_Number:** KDB-7000-B2-REP**
- ****Maintenance_Schedule:** 250-hour intervals for inspection; replacement at 2,000 cumulative operating hours or when wear exceeds 0.020 inches**

SECTION 1 - USE CASE

- Provides rotational interface between kelly and drive motor with minimal friction loss
- Accommodates vertical and lateral kelly movement during drilling operations
- Contains roller bearing assembly to support radial loads up to 45,000 pounds
- Enables quick-disconnect coupling for rapid kelly changes during tripping operations
- Maintains concentricity tolerance of ± 0.005 inches for optimal drill string alignment

SECTION 2 - FAILURE CAUSES

- ****E-501 (Bushing Wear Detected)**** – Abrasive drilling fluid particles causing internal surface erosion; frequency: 28% of failures; accelerated by high-viscosity muds
- ****E-502 (Roller Bearing Seizure)**** – Inadequate lubrication or contaminated grease; frequency: 22% of failures; results in complete rotational lockup
- ****E-503 (Lateral Play Excessive)**** – Bearing preload loss from thermal cycling; frequency: 19% of failures; causes vibration and kelly misalignment
- ****Seal Failure**** – Drilling fluid ingress into bearing cavity; frequency: 18% of failures; leads to corrosion and bearing degradation
- ****Coupling Pin Shear**** – Sudden torque spikes from stuck pipe; frequency: 13% of failures; requires immediate kelly disconnection

SECTION 3 - SOLUTIONS

- **E-501 Response:** Measure bushing inner diameter at three axial locations; if wear exceeds 0.020 inches, schedule replacement during next planned maintenance window; implement mud filtration upgrade (API RP 13C) to reduce particle size
- **E-502 Response:** Execute controlled shutdown per OSHA 1910.147; disassemble bushing assembly; inspect roller bearings for discoloration or spalling; replace bearing set with API-certified components; relubricate with NLGI Grade 2 synthetic grease
- **E-503 Response:** Measure lateral play using dial indicator at kelly top; if exceeds 0.010 inches, adjust bearing preload per manufacturer specifications; if adjustment insufficient, replace bushing assembly
- **Seal Replacement:** Remove kelly from drive bushing; drain bearing cavity; replace elastomer seals and backup rings; apply fresh lubricant; pressure test seal integrity at 1.25x operating pressure
- **Coupling Pin Inspection:** After any stuck pipe incident, visually inspect coupling pins for cracks or deformation; replace if any damage detected; verify pin torque specification (180-220 ft-lbs) per API RP 54 Section 6.4

SECTION 4 - INVENTORY DATA

- **Current Stock:** 4 complete KDB-7000-B2 units; 16 roller bearing assemblies; 20 seal kits; 24 coupling pins
- **Reorder Threshold:** Maintain minimum 2 complete units and 12 seal kits; trigger reorder when stock reaches threshold
- **Supplier:** PrecisionBore Manufacturing Corp., Corpus Christi TX; secondary supplier: Gulf Coast Drilling Components, Galveston TX (lead time 16 days)
- **Lead Time & Cost:** Primary supplier 10 business days; cost per unit \$28,900 USD; expedited 5-day delivery available (+\$2,100)

UNIT 3: ROTARY SWIVEL ASSEMBLY

Unit_Name: Rotary Swivel with Pressure Bearing
Unit_Id: RSA-7000-C3
Manufacture_Date: 2023-03-30
Manufacturer_Company_Name: SwiftFlow Hydraulic Systems
Manufacturer_Company_Id: SFS-HYD-4156
Inventory_Quantity: 3 units in stock
Supplier_Information: SwiftFlow Hydraulic Systems, Lafayette LA; Contact: technical@swiftflowhydraulic.com; Lead Time: 12 business days

DOMAIN SPECIFICATIONS

- **Wellbore_Depth_Rating:** 20,000 feet (6,096 meters)
- **Operating_Pressure_PSI:** 4,500 PSI maximum continuous; 6,000 PSI rated for pressure bearing
- **Drilling_Fluid_Compatibility:** Water-based and synthetic muds; rated for mud weights 8.5-18.0 ppg
- **Known_Error_Codes:** E-601 (Pressure Bearing Leak), E-602 (Swivel Rotation Resistance), E-603 (Mud Inlet Blockage)

- **Replacement_Part_Number:** RSA-7000-C3-REP
- **Maintenance_Schedule:** 300-hour intervals for seal inspection; bearing replacement at 3,000 cumulative hours or upon pressure test failure

SECTION 1 - USE CASE

- Provides rotating mud inlet connection between standpipe and kelly while supporting full drilling fluid pressure
- Accommodates vertical kelly movement (up to 40 feet) without interrupting mud circulation
- Pressure bearing supports axial loads from kelly weight and drill string tension (up to 500,000 pounds)
- Enables continuous drilling fluid delivery at rates up to 1,200 gallons per minute
- Isolates rotating drill string from stationary surface equipment via sealed bearing interface

SECTION 2 - FAILURE CAUSES

- **E-601 (Pressure Bearing Leak)** – Seal degradation from thermal cycling or abrasive mud particles; frequency: 26% of failures; manifests as mud weeping at bearing housing
- **E-602 (Swivel Rotation Resistance)** – Bearing preload excessive or bearing corrosion; frequency: 21% of failures; increases motor load and reduces drilling efficiency
- **E-603 (Mud Inlet Blockage)** – Mud solids accumulation or scale formation in inlet passages; frequency: 20% of failures; reduces flow rate and increases back-pressure
- **Thrust Bearing Wear** – Inadequate lubrication or contaminated bearing grease; frequency: 18% of failures; causes axial play and kelly instability
- **Swivel Housing Crack** – Fatigue from cyclic pressure loading; frequency: 15% of failures; requires immediate replacement to prevent catastrophic failure

SECTION 3 - SOLUTIONS

- **E-601 Response:** Reduce mud circulation pressure by 10%; inspect seal condition through observation port; if weeping continues, schedule swivel replacement during next tripping operation; apply temporary sealant (API-approved) only as emergency measure; perform full seal replacement per API RP 54 Section 10.2
- **E-602 Response:** Measure swivel rotation torque using calibrated torque wrench; if exceeds 150 ft-lbs, reduce bearing preload per manufacturer manual (typically 0.5-turn counterclockwise); if resistance persists, disassemble and inspect bearing for corrosion; replace bearing assembly if pitting detected
- **E-603 Response:** Cease drilling operations; flush inlet passages using clean drilling fluid at low pressure (500 PSI); if blockage persists, disassemble inlet manifold and mechanically clear obstruction; implement mud filtration upgrade (API RP 13C) to prevent recurrence
- **Bearing Lubrication:** Apply ISO VG 32 synthetic bearing grease every 200 operating hours; use grease gun with pressure relief to prevent over-pressurization; inspect grease color for contamination (dark brown indicates mud ingress)
- **Pressure Test Protocol:** After any seal replacement or bearing service, perform hydrostatic pressure test at 1.5x maximum operating pressure (6,750 PSI) for minimum 10 minutes; monitor for any pressure drop; document results per OSHA 1910.119 Process Safety Management

SECTION 4 - INVENTORY DATA

- **Current Stock:** 3 complete RSA-7000-C3 units; 12 pressure bearing assemblies; 15 seal kits; 6 inlet manifold assemblies
- **Reorder Threshold:** Maintain minimum 1 complete unit and 10 seal kits; trigger reorder when stock reaches threshold
- **Supplier:** SwiftFlow Hydraulic Systems, Lafayette LA; secondary supplier: Bayou Drilling Equipment Inc., Morgan City LA (lead time 18 days)
- **Lead Time & Cost:** Primary supplier 12 business days; cost per unit \$62,400 USD; emergency 3-day expedited service available (+\$5,800)

COMPLIANCE SUMMARY

All procedures documented herein comply with:

- **OSHA 1910.147** – Lockout/Tagout requirements for maintenance operations
- **OSHA 1910.303** – Electrical safety standards for connector maintenance
- **OSHA 1910.119** – Process Safety Management documentation requirements
- **API RP 54** – Recommended Practice for Occupational Safety and Health for Onshore Oil and Gas Production Operations
- **API RP 13B-1** – Recommended Practice for Field Testing Water-Based Drilling Fluids
- **API RP 13C** – Recommended Practice for Drilling Fluids Processing Systems Evaluation

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Authorized By: Regional Operations Manager, Meridian Offshore Drilling Solutions LLC

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